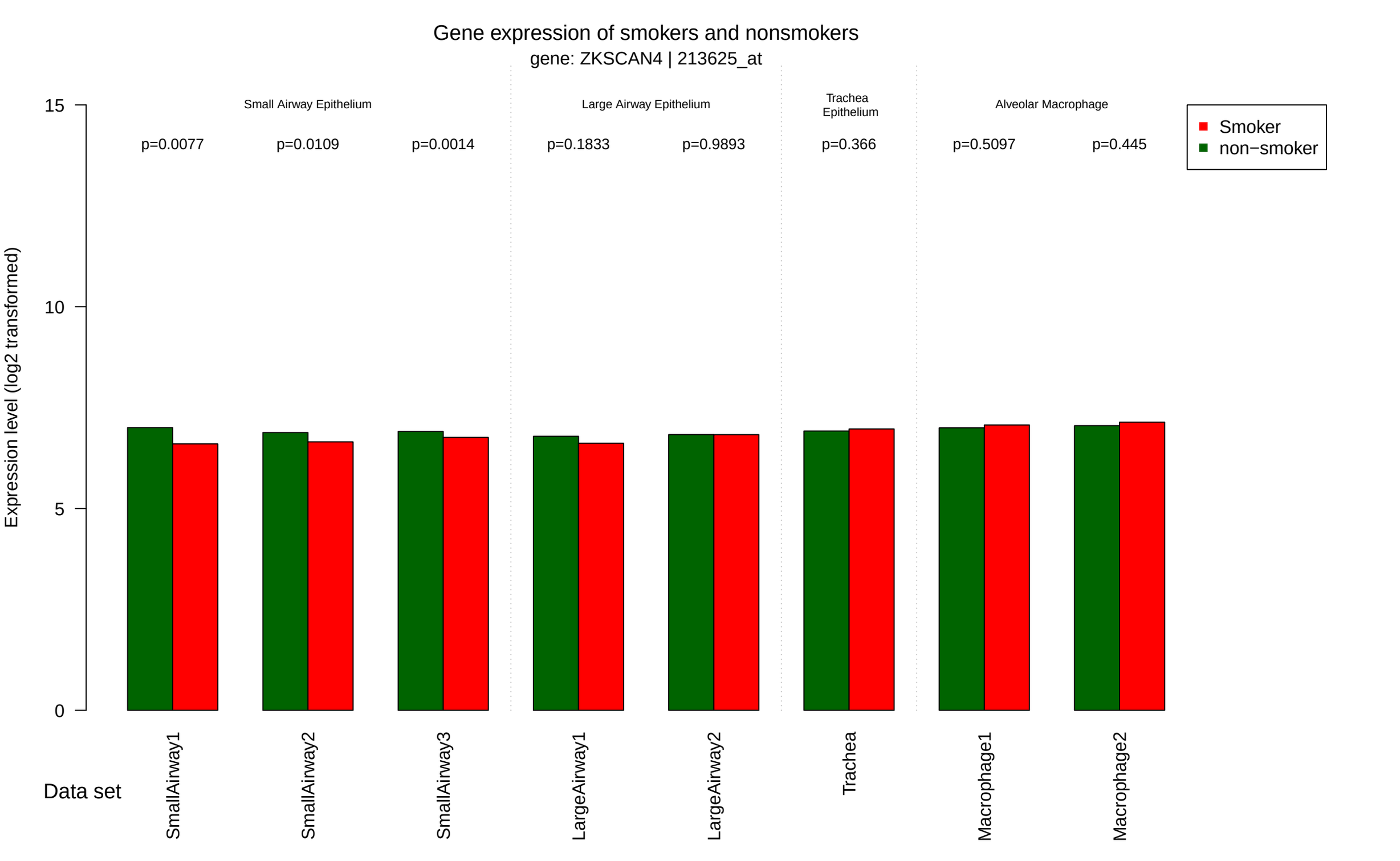
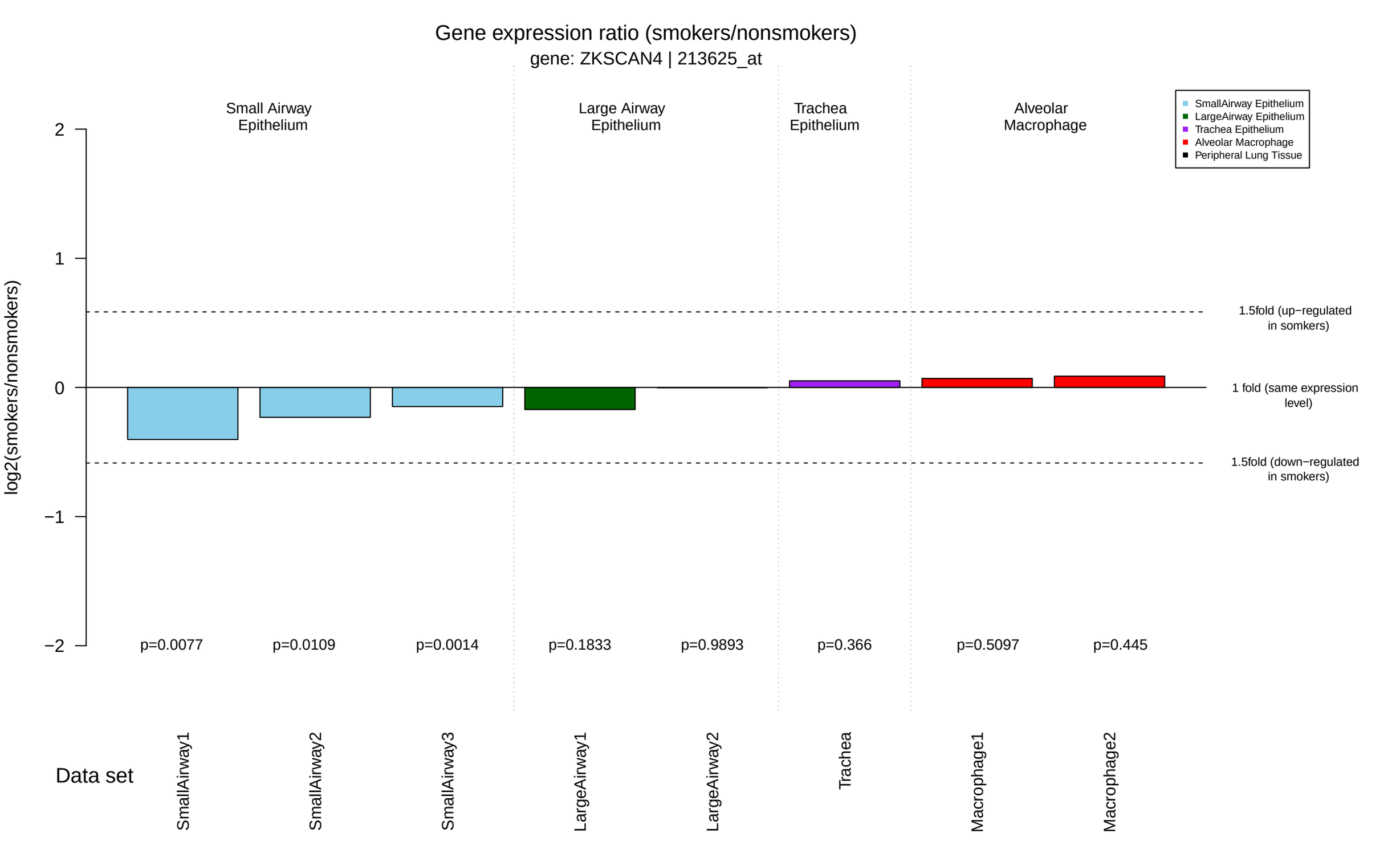
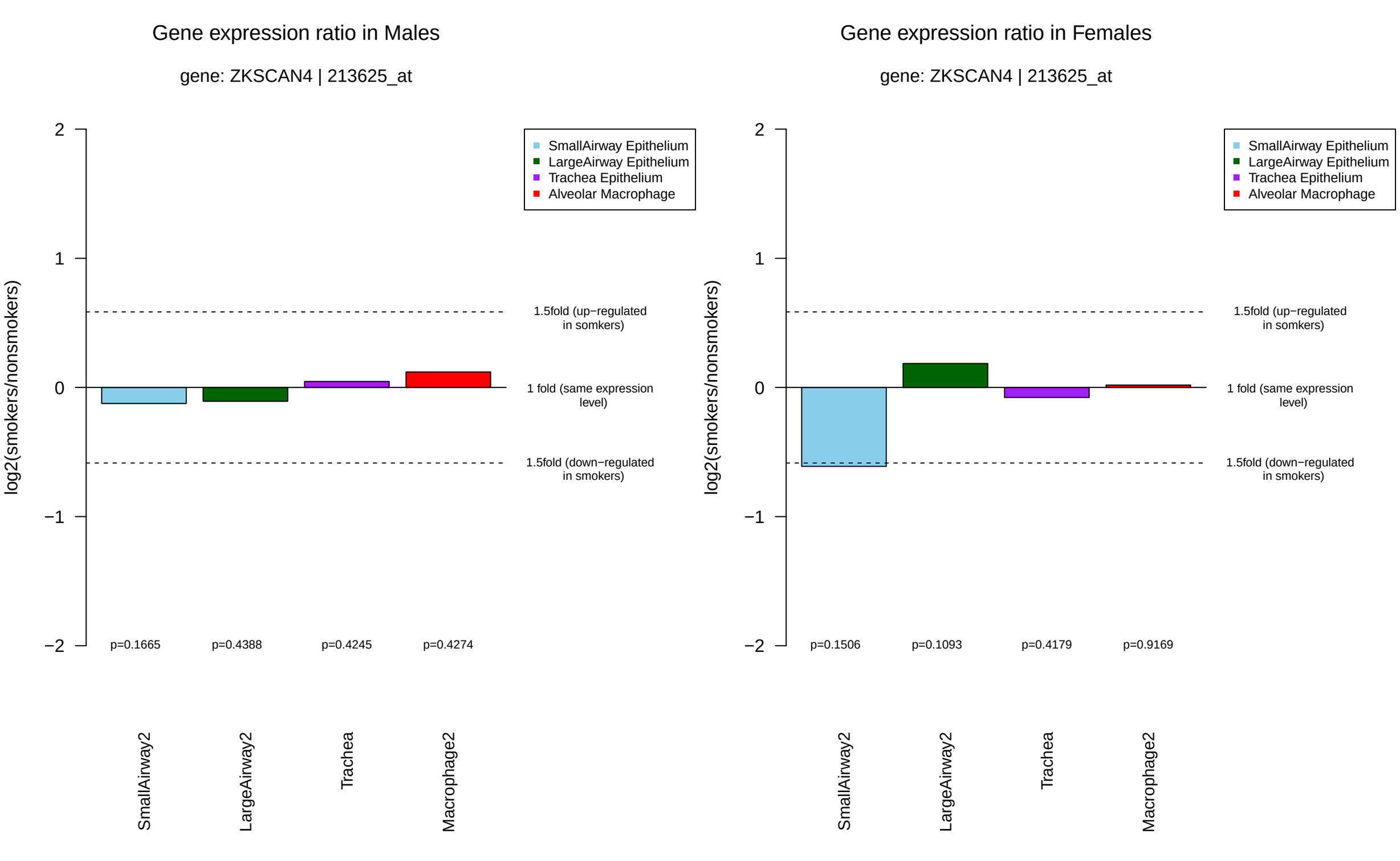
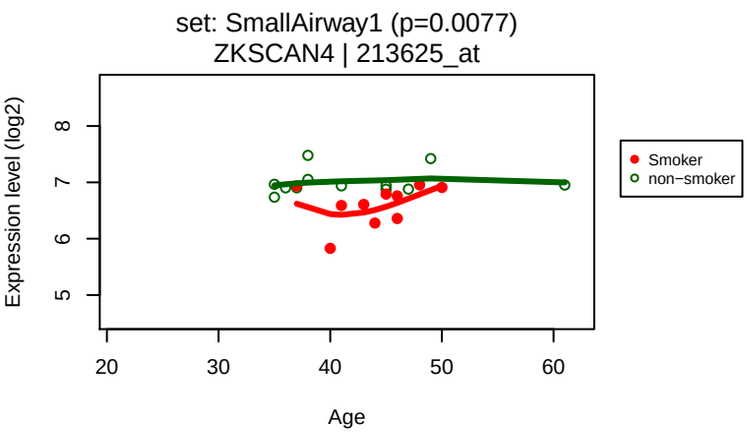
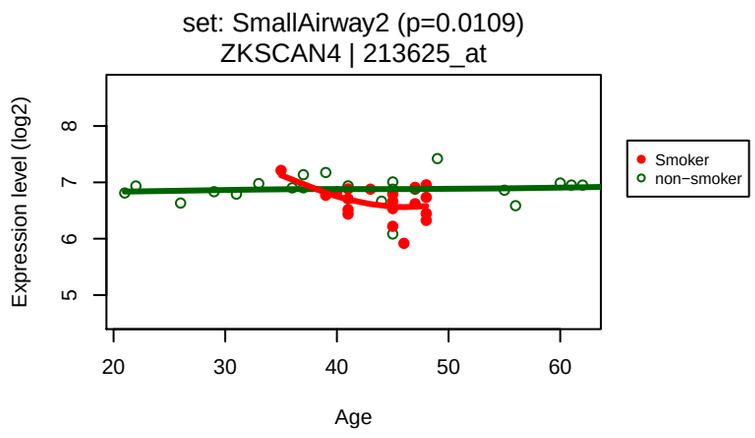
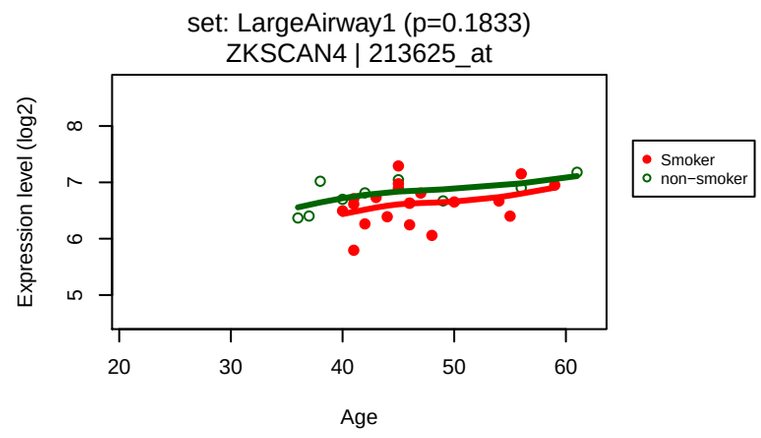

Gene expression of smokers and nonsmokers
gene: ZKSCAN4 | 213625_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0077
p=0.0109
p=0.0014
p=0.1833
p=0.9893
p=0.366
p=0.5097
p=0.445
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

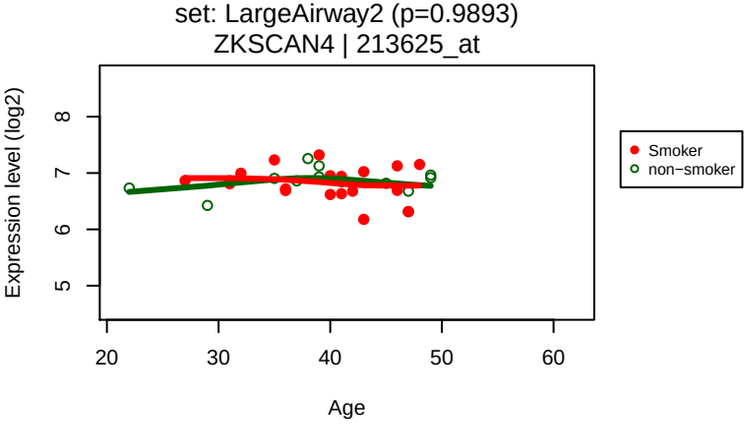

Gene expression ratio (smokers/nonsmokers)
gene: ZKSCAN4 | 213625_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0077
p=0.0109
p=0.0014
p=0.1833
p=0.9893
p=0.366
p=0.5097
p=0.445
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

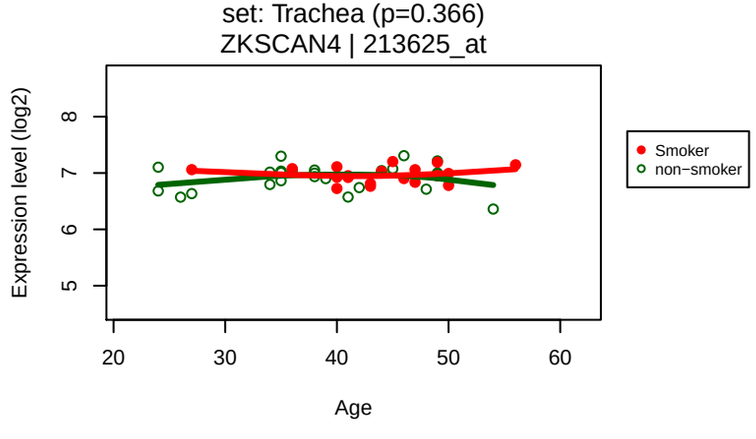

Gene expression ratio in Males
gene: ZKSCAN4 | 213625_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1665
p=0.4388
p=0.4245
p=0.4274
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ZKSCAN4 | 213625_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1506
p=0.1093
p=0.4179
p=0.9169
SmallAirway2
LargeAirway2
Trachea
Macrophage2

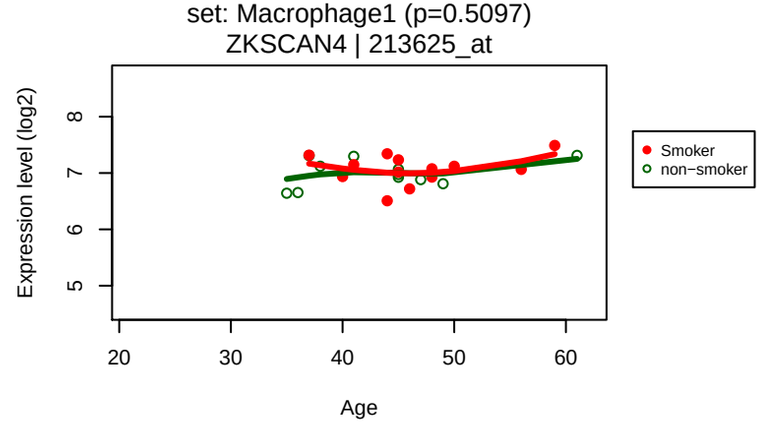

set: SmallAirway1 (p=0.0077)
ZKSCAN4 | 213625_at
Expression level (log2)
Age
Smoker
non−smoker

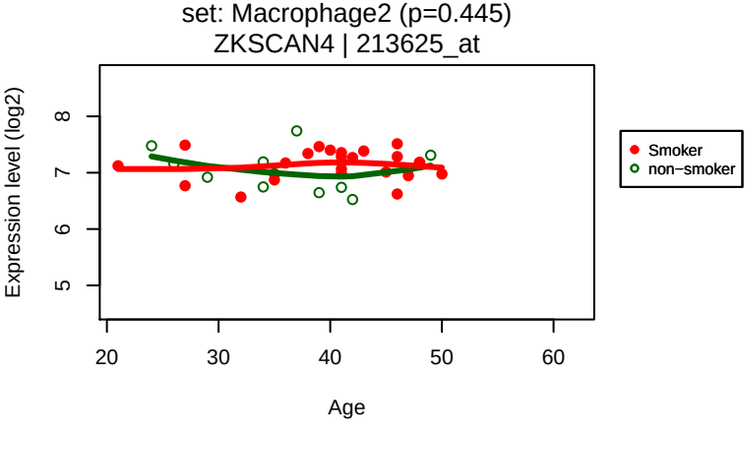

set: SmallAirway2 (p=0.0109)
ZKSCAN4 | 213625_at
Expression level (log2)
Age
Smoker
non−smoker

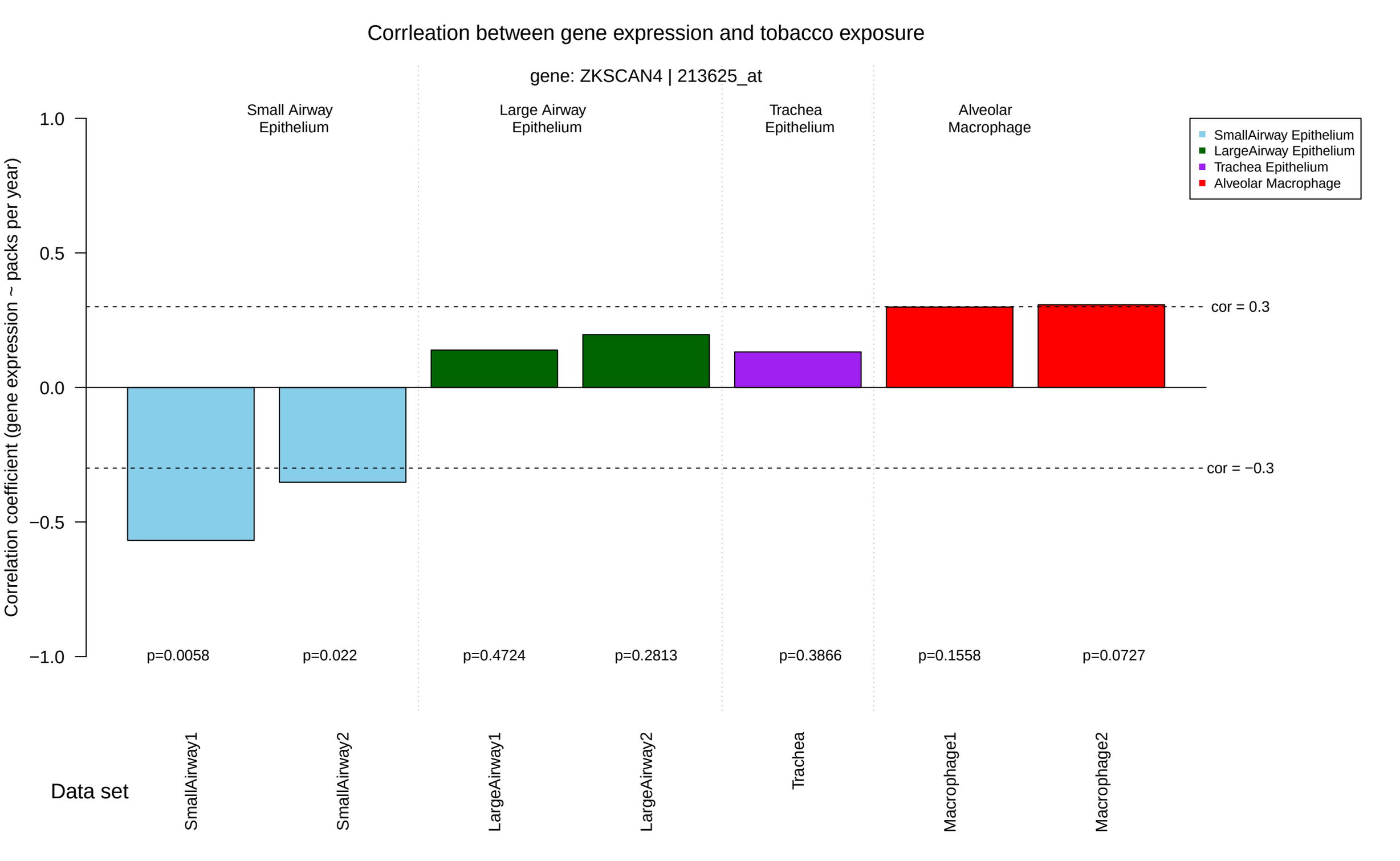

set: LargeAirway1 (p=0.1833)
ZKSCAN4 | 213625_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9893)
ZKSCAN4 | 213625_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.366)
ZKSCAN4 | 213625_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5097)
ZKSCAN4 | 213625_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.445)
ZKSCAN4 | 213625_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ZKSCAN4 | 213625_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0058
p=0.022
p=0.4724
p=0.2813
p=0.3866
p=0.1558
p=0.0727
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2